INVESTIGATION REPORT

INV-2024-0067

NEXUS Assembly Line Mix-up Investigation

Investigation ID:	INV-2024-0067
Product:	NEXUS Troponin (Reference 30450)
Status:	Closed
Date Opened:	2024-03-10
Date Closed:	2024-05-15
Lead Investigator:	L. Reynolds - Quality Engineering
Approved By:	P. Dawson - Quality Director
Classification:	Major - Product recall (1 lot)

1. Executive Summary

Investigation INV-2024-0067 was initiated following a customer complaint reporting that NEXUS Troponin kit lot 1010550120 contained sensor cartridges labelled for NEXUS CK MB. The customer identified the issue during routine kit inspection before use. No patient results were affected.

The investigation determined that the root cause was a failure to follow the assembly line changeover procedure (SOP-ASM-003). NEXUS CK MB production was followed by NEXUS Troponin on assembly line AL-003, and the changeover was completed in only 15 minutes instead of the mandated 30-minute minimum. Residual CK MB CART cartridges were not cleared from the feed hopper, resulting in their inclusion in Troponin kits.

2. Problem Description (QQOCQPC)

Qui (Who):
Customer site - Regional Medical Center. Assembly operator OP-045 (J. Mitchell) performed the changeover on assembly line AL-003.

Quoi (What):
NEXUS CK MB sensor cartridges (CART-CKMB-001) found in NEXUS Troponin kit lot 1010550120. Approximately 48 kits in the lot may be affected (first production run after changeover).

Ou (Where):
Assembly line AL-003, Building B, Riverside facility. Customer complaint originated from Regional Medical Center, Riverside.

Comment (How):
During changeover from CK MB to Troponin production, the operator did not fully clear the sensor cartridge feed hopper. Residual CK MB cartridges were mixed with incoming Troponin cartridges. Barcode scanner was in bypass mode due to an earlier scanner fault (STOP-2024-0015).

Quand (When):
Changeover occurred on 2024-03-05 at 14:30. Production of Troponin lot 1010550120 started at 14:45 (only 15 minutes after changeover vs. 30-minute minimum per SOP-ASM-003). Customer complaint received: 2024-03-10.

Pourquoi (Why):
Operator rushed the changeover due to production schedule pressure. The 8-item clearance checklist was signed but items 3 (hopper clearance) and 6 (barcode scanner verification) were not actually performed.

Combien (How much):
1 lot affected (1010550120, ~2,500 kits). Voluntary recall of entire lot. Estimated cost: 45,000 EUR (recall + investigation + replacement production). No patient harm reported.

3. Data Analysis

3.1 Production Chronology

Lot	Product	Start Time	End Time	Status	Notes
1010550115	CK MB	2024-03-04 08:00	2024-03-04 16:30	Released	Normal production
1010550118	CK MB	2024-03-05 08:00	2024-03-05 14:15	Released	Last CK MB lot before changeover
--	Changeover	2024-03-05 14:15	2024-03-05 14:30	--	ONLY 15 MIN (SOP requires 30 min)
1010550120	Troponin	2024-03-05 14:45	2024-03-05 22:00	RECALLED	First Troponin lot - AFFECTED
1010550121	Troponin	2024-03-06 08:00	2024-03-06 16:30	Released	No issues - hopper fully replenished

3.2 Changeover Analysis

Item	Checklist Step	Operator Signature	Verification Result
1	Previous product materials removed from line	Signed	Compliant
2	Feed conveyors emptied and cleaned	Signed	Compliant
3	sensor cartridge hopper cleared and verified empty	Signed	NOT PERFORMED
4	reagent card magazine replaced	Signed	Compliant
5	Label rolls changed and verified	Signed	Compliant
6	Barcode scanner operational check	Signed	NOT PERFORMED (scanner in bypass)
7	First article inspection completed	Signed	Compliant
8	Supervisor sign-off	Not signed	NOT PERFORMED

3.3 Equipment Review

Assembly line AL-003 mechanical inspection showed no equipment faults. However, the barcode scanner (BC-003) was in bypass mode since 2024-03-03 due to a communication fault (STOP-2024-0015). A repair request had been submitted but was pending maintenance scheduling. With the scanner in bypass, automatic rejection of mismatched sensor cartridges was disabled.

4. Root Cause Analysis

Root Cause: Changeover procedure (SOP-ASM-003) not followed. Operator did not clear sensor cartridge hopper and did not verify barcode scanner operation. Changeover completed in 15 minutes vs. 30-minute minimum requirement.
5M Category: Methode (Method) / Process
Contributing Factors:
- Production schedule pressure: two product changeovers in one day.
- Barcode scanner in bypass mode (STOP-2024-0015), disabling automatic detection.
- Supervisor sign-off (checklist item 8) was not enforced.
- No physical interlock to prevent production start with incomplete changeover.

5. Conclusions and Corrective Actions

Conclusion: The assembly line mix-up was caused by failure to follow the changeover procedure. The barcode scanner bypass removed the last automated safety barrier. Enhanced procedural controls and equipment interlocks are required to prevent recurrence.

CAPA ID	Description	Type	Due Date	Status
CAPA-2024-0034	Implement electronic changeover timer - production cannot start until 30 minutes have elapsed	Corrective	2024-06-01	Completed
CAPA-2024-0035	Add interlock: production blocked if barcode scanner is in bypass mode	Corrective	2024-06-01	Completed
CAPA-2024-0036	Mandatory supervisor physical presence for changeover sign-off (item 8 cannot be remote-signed)	Corrective	2024-04-15	Completed
CAPA-2024-0037	Retrain all assembly operators on SOP-ASM-003 with emphasis on hopper clearance verification	Preventive	2024-04-30	Completed
CAPA-2024-0038	Add second independent hopper inspection (different operator) to changeover checklist	Preventive	2024-05-15	Completed

6. Effectiveness Verification

Post-CAPA monitoring (June - August 2024): 12 changeovers performed on AL-003 with new electronic timer and interlock. All changeovers met the 30-minute minimum. Barcode scanner uptime maintained at 99.8%. No recurrence of component mix-up. Zero customer complaints for products assembled on AL-003 during monitoring period. CAPAs verified effective and investigation closed on 2024-05-15.